# UAV Micro-Doppler Signature Analysis Using DVB-S Based Passive Radar

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***Abstract*—Drones and unmanned aerial vehicles (UAVs) are increasingly popular, thus posing danger and threats to infrastructures and public safety. A technology for drone detection and classification would therefore significantly increase the level of security. In scenarios such as concerts, sport events, trade fairs, or in any situation where significant aggregation of people is present, such techniques should be non-invasive. That means they do not have to pose an additional threat to people themselves. To this end, passive radars offer an appealing solution, since they are able to offer a non-cooperative surveillance while not emitting any electromagnetic signal. On the contrary, they rely on existing transmitting infrastructure (also referred to as illuminators of opportunity, IoO), such as broadcasting signal sources (FM radio, terrestrial and satellite digital video broadcasting, cellular communication and so on). In this work, the possibility to exploit satellite television based passive radar for UAV detection is analyzed by experimental validation. In addition, micro-Doppler signatures for drones have been extracted, which might give information for subsequent UAV classification.**

## I. Introduction

Passive bistatic radar (PBR) represents a unique class of radar sensors, that lack a dedicated transmitter. PBR relies instead on so called illuminators of opportunity (IoO) which are not under the control of the radar operator. Some of the most commonly exploited transmission sources are public terrestrial broadcasting and communication signals such as cellular networks. A primary advantage of such systems is that no electromagnetic spectrum needs to be allocated in order for them to operate. Without their own signal emissions these system represent a sensing option for highly covert surveillance.

Although passive radars cannot fully substitute active radar in all deployment scenarios, there are specific applications in which their covert operation capability is a key enabler characteristic.

Possible applications include the protection of critical infrastructure such as airports, harbors, power plants and so on, against the intrusion of small unmanned aerial vehicles (UAV). The ability of PBR systems to detect small UAV has been demonstrated with a multitude of different terrestrial IoO such as cellular communication signals [1], digital and analog radio [2], [3], as well as television broadcasting [4], [5], [6], [7]. The performance of such systems however depend heavily on the availability of suitable IoOs, that provide sufficient bandwidth and signal power density.

In contrast, satellite based IoOs offer ground coverage with a much larger footprint and a more evenly distributed power density.

Earlier work focused on the illumination from synthetic aperture radar (SAR) satellites in a bistatic configuration with a stationary ground based receiver [8]. The potential of using signals from global navigation satellite systems (GNSS) for the detection of maritime targets [9], [10], [11], ground moving targets [12] and UAVs [13] has been demonstrated recently.

These IoO have in common, that they are mounted on platforms in the low or medium earth orbit, which means that they present non-stationary sources of illumination. PBR systems need to compensate accordingly, thus leading to higher system complexity and reduced reliability [14]. In contrast geostationary satellites represent illuminators with negligible relative velocity with respect to the receiver. Recent work has shown that transmissions from these kinds of spaceborne platforms can be exploited for bistatic radar detection [15] and imaging tasks of maritime targets [16].

Within the scope of this paper, UAV detection based on the exploitation of communication signals from geostationary platforms is investigated. More specifically transmissions modulated according to the second generation digital video broadcasting standard for satellite (DVB-S2) [17], which represents one of the most widely available transmission scheme for satellite based audio and video distribution.

These kinds of broadcast services are designed for constant wide area coverage with high bandwidths. However, transmissions suffer from low signal spectral density at Earth's surface. To cope with this drawback, antennas with high directivity on receive are necessary in order to guarantee an adequate link budget [18]. The aim is to ensure reliable detection performance on short to medium ranges for UAV early warning applications.

This work presents the first results obtained for UAV detection, where a pre-existing PBR demonstrator designed for the utilization of satellite broadcasting transmissions [19] is used. Specifically, it will be shown that UAV detection is possible, and that micro-Doppler signatures for UAVs are also visible in the range-Doppler map. This micro-Doppler signature, which requires further and much deeper analysis, is generated by the rotating blades of the UAV's propellers, and it might help in classifying the UAV.

The remainder of this paper is organized as follows. In Section II the operational principle of a PBR is briefly recalled

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and the experimental setup which has been used to prove UAV detection capability is reported in Section III. Section IV describes the data post processing scheme that was used to isolate the UAV's micro-Doppler signatures. Finally, Section V shows the first results obtained with a short outlook of future work in Section VI.

## II. Basic Passive Radar Principle and Processing

The PBR setup used in these measurements consisted of two receiving channels, each one connected to a separate antenna. The antenna connected to the reference channel points towards the satellite in order to get a clear (reference) signal, while the second one points a specific area to be surveilled. The geometry of the system is illustrated in Fig. 1.

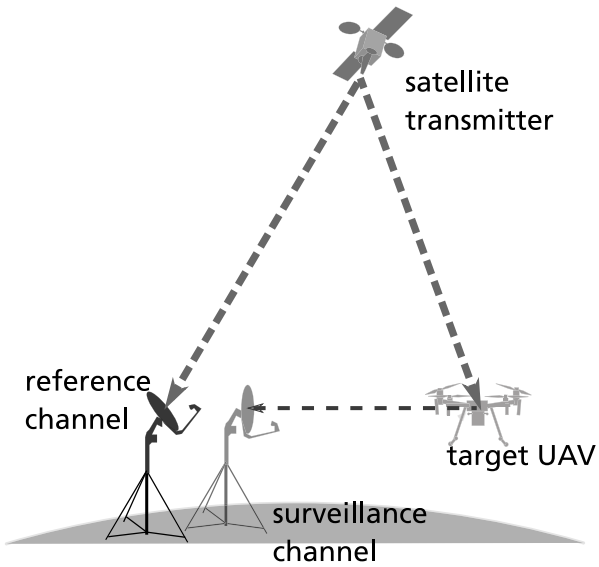


Fig. 1: Satellite based passive radar geometry.

The working principle of a passive radar is based on processing the signals received from the two channels. Specifically, after reception, down-conversion and digitization, these two signals are cross-correlated for varying potential Doppler shifts of the reference signal, thus calculating a cross-ambiguity function (CAF). The result of the CAF is usually depicted in a range-Doppler map, where target detection stage is then performed.

The expression of the CAF is as follows:

$$\chi(\tau, f_D) = \int_{-\infty}^{+\infty} s_{surv}(t) \cdot s^*_{ref}(t-\tau) \cdot e^{-j2\pi f_D t} dt \quad (1)$$

where $s_{surv}$ is the complex signal received by the surveillance channel and $s_{ref}$ the reference signal, * denotes the complex conjugate, $\tau$ is the time difference of arrival (TDOA), and $f_D$ is the potential Doppler shift. The temporal integration is limited by the coherent time interval. The ambiguity function corresponds to the response of a matched filter shifted in Doppler. The processing is summarized in the block diagram in Fig. 2, where the final processing stage aiming at analyzing the micro-Doppler signature of detected targets is reported as well.

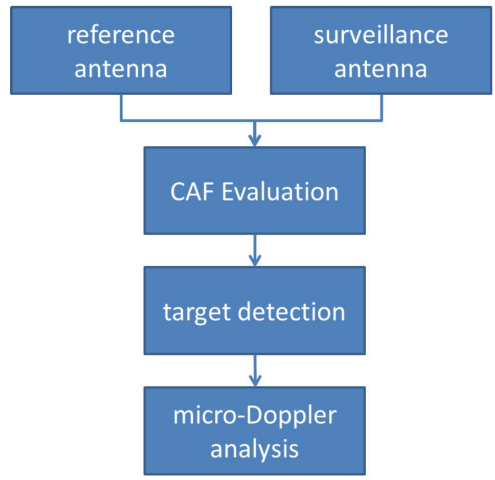


Fig. 2: Passive radar signal processing chain.

## III. Experimental Setup

The experiments have been carried out with the satellite based passive radar system SABBIA, which was previously used for inverse synthetic aperture radar (ISAR) applications [16], [20].

SABBIA's antenna configuration consists of two identical commercial off-the-shelf (COTS) parabolic dish antennas, which were designed for a frequency range from 10.75 GHz up to 12.75 GHz, providing a gain between 38.12 dBi and 39.53 dBi. The elements were mounted on two separate tripods, each equipped with antenna positioner that allowed the elements to be independently steered in elevation and azimuth. The antennas orientation was tracked and logged with an attached internal measurement units (IMU).

The signals impinging on each antenna dish are reflected towards custom designed antenna horns at an offset angle of 22.5° and passed to a Quad-LNB (low noise block). The block diagram in Fig. 3 shows how the LNBs perform the first stage of amplification and down-conversion of the signal to 950-1950 MHz (DVB-S low-band) and 1100-2150 MHz (DVB-S high-band), respectively.

An external 10 MHz low phase-noise oscillator synchronizes

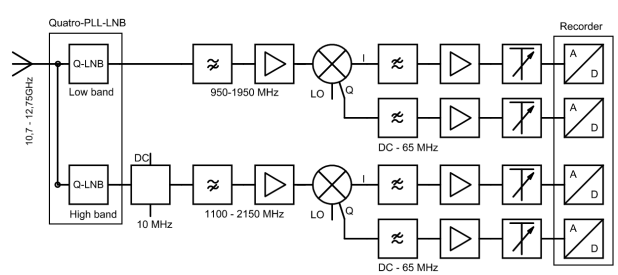


Fig. 3: SABBIA front end receiver block diagram.



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the receiving elements to ensure phase coherence between the channels. The subsequent filtering and amplification stages are performed with FHR's custom build ATLAS receiver [21], were the signal is converted further down and split into their seperate I/Q-components. A combination of amplifiers and digitally adjustable attenuators ensures an ideal level of the signals prior to analog-to-digital sampling.

For the trial campaign, the ATLAS front end was configured to filter, down convert and sample 80 MHz of instantaneous signal bandwidth within the $K_u$ spectrum transmitted by the communication satellite. The geostationary satellite Astra 1KR [22] was selected as IoO for these trials. It is part of the ASTRA 19.2E group of co-located communication satellites that provide service across Europe with high definition television down link using the DVB-S2 standard. Information about the satellite position and transmission parameters are listed in Table I.

For these measurements downlink data was captured simultaneously in vertical (V-pol.) and horizontal (H-pol.) polarization around ASTRA 1KR's transponder No. 10 at a center frequency of 11.347 GHz. Because of the limited bandwidth of the ATLAS receiver only a fracture of the total 1 GHz satellite downlink spectrum could be recorded. System performance in terms of range resolution and SNR could therefore be potentially increased with a higher bandwidth receiver.

TABLE I: Satellite Illuminator

| Name | Astra 1KR |
|---|---|
| Longitude | 19.2° |
| Inclination | 0.06° |
| Perigee altitude | 35,774 km |
| Apogee altitude | 35,811 km |
| EIRP | 51 dBW |
| Downlink bandwidth | 10.7 - 11.7 GHz |
| Number of transponder | 32 H-Pol., 32 V-Pol. |

The passive radar system was deployed as shown in the illustration in Fig. 1, namely with one reference antenna pointing at the satellite IoO (Tx), and the surveillance antenna pointing at the area where flights with the UAV were performed, see Fig. 4. Two different types of small commercial UAVs have been considered:

- a quadcopter F-450 (see Fig. 5 (a));
- a hexacopter F-550 (see Fig. 5 (b)).

Their specific technical characteristics are listed in Table II. The primary aim of this experiment was to investigate whether the passive radar system could detect the UAVs and possibly also the micro-Doppler signature of their rotors.

## IV. Data Processing

Data has been recorded from the two channels, down-converted to base band and processed according to (1) and to the processing scheme in Fig. 2. The recorded DVB-S2 signal is transmitted as raised-cosine filtered continuous waveform (CW). CAFs were calculated from the received down convert

TABLE II: UAV Technical Characteristics

| Name | F-450 | F-550 |
|---|---|---|
| Manufacturer | DJI | DJI |
| Empty Weight | 2 Kg | 3 Kg |
| Maximum Take-off-weight | 3 Kg | 5 Kg |
| Number of Rotors | 4 | 6 |
| RPM | 5000-7000 $\mathrm{min}^{-1}$ | 5000-7000 $\mathrm{min}^{-1}$ |
| Rotor Diameter | 27.9 cm | 25.4 cm |
| Dimensions (length x width) | 50 x 50 cm | 60 x 60 cm |

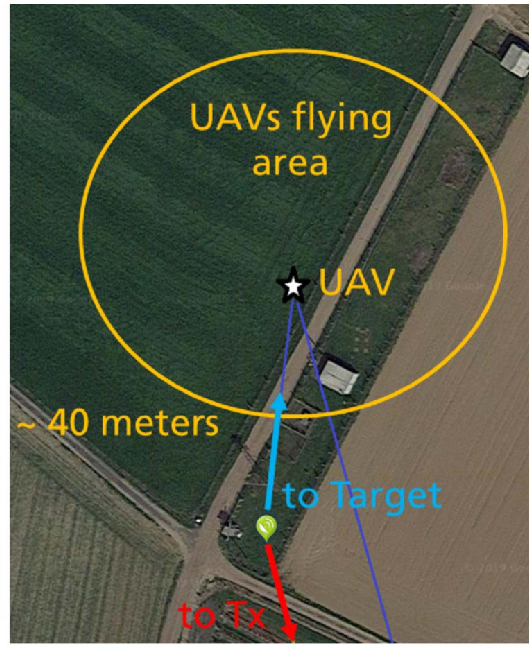


Fig. 4: Aerial image of the geometry used during the experiment.

Fig. 5: Quadcopter F-450 (a), Hexacopter F-550 (b).

80 MHz signals $s_{ref}$ and $s_{surv}$ with 419 ms of coherent integration time in order to achieve sufficient processing gain for detection. Range profiles with short maximum unambiguous ranges of 7.67 km were chosen in order to improve Doppler-resolution for the micro-Doppler analysis. Thus the CAF was formed from 16384 range-profiles which gave a Doppler-resolution of 2.38 Hz.



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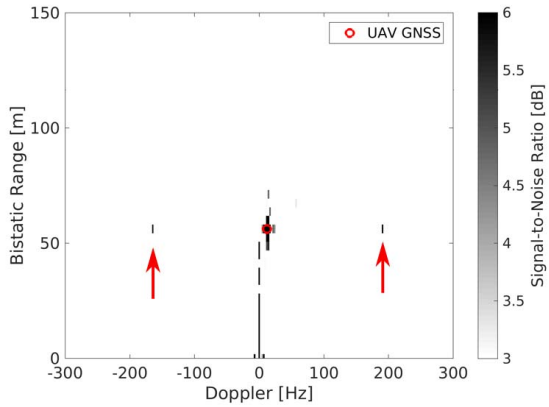
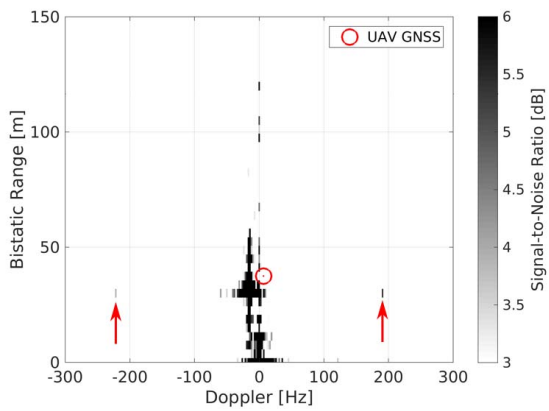
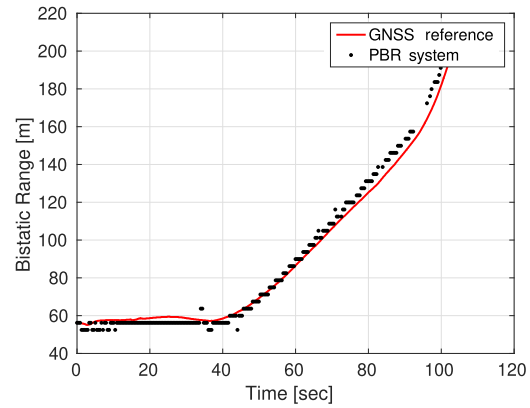



(a) Quadcopter F-450

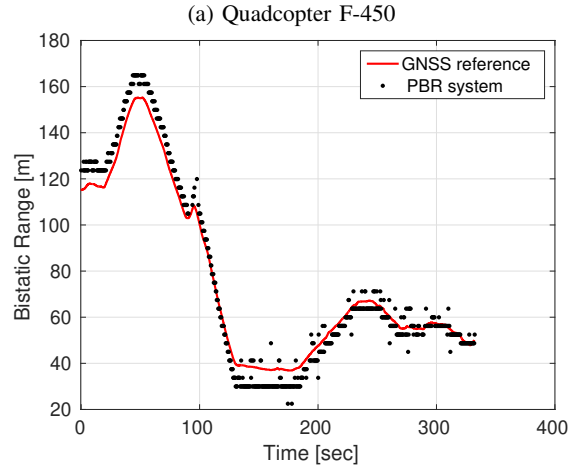



(b) Hexacopter F-550

Fig. 6: Range-Doppler maps with UAV detection. Red arrows indicate potential micro-Doppler signatures.




(a) Quadcopter F-450




(b) Hexacopter F-550

Fig. 7: Bistatic range measurements.

Two snap shots of the resulting CAF are shown in Fig. 6a and in Fig. 6b for the F-450 and F-550, respectively. As is apparent, the echo coming from the body of the UAVs coincide with secondary Doppler-shifted returns within the same bistatic range (highlighted with red arrows). The close to ideal thumb-tack response of the CAF allowed to set a constant threshold of 3 dB above the noise floor. This resulted in a tolerable false alarm rate of approximately $0.5 \cdot 10^{-6}$.

Ground-truth trajectories in the bistatic range and Doppler domain were also calculated based on GNSS recordings on-board of both UAVs. CAF detections were associated to these trajectory points if they fell within a window of 40 Hz in Doppler and 15 m in bistatic range. A comparison of range accuracy between ground truth and passive radar data is presented in Fig. 7. The corresponding trajectory in the Cartesian domain is displayed in Fig. 8.

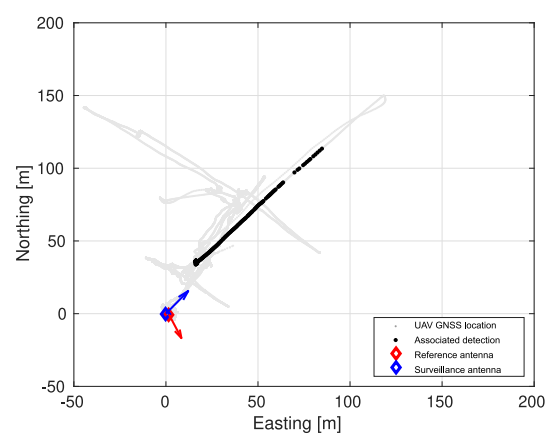



Fig. 8: Cartesian trajectory of the F-450 UAV

## V. Results

The verification of the UAV trajectories with the IMU reference, allowed for the extraction of Doppler-profiles associated to the UAV. Fig. 9 shows 120 consecutive profiles of the F-450 and 70 profiles for the F-550. These plots shows, that the UAV's estimated Doppler shift obtained from the GNSS reference is well aligned with the PBR's measurements. In addition two parallel Doppler shifted signatures are also visible.



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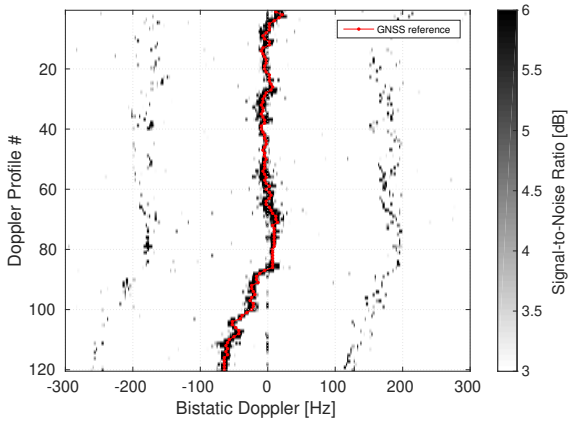
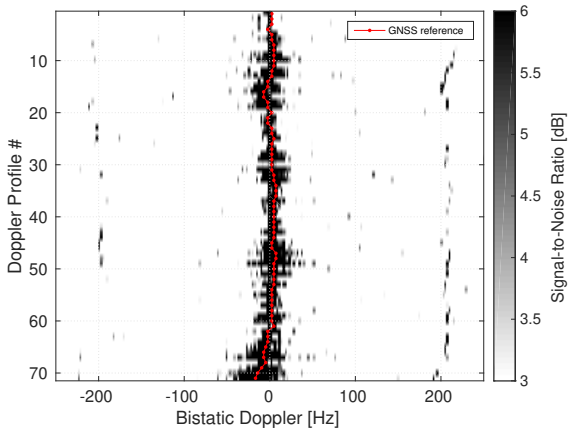
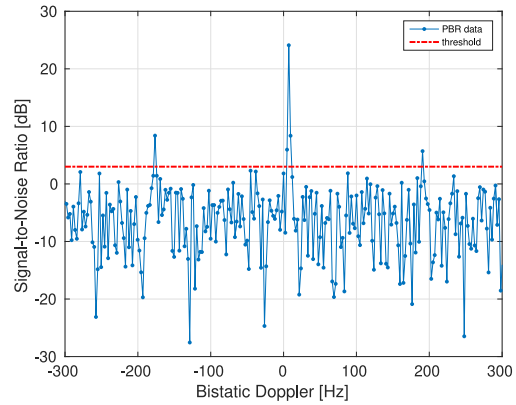



(a) Quadcopter F-450

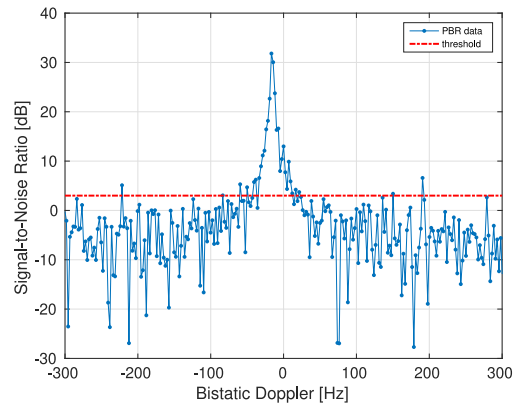



(b) Hexacopter F-550

Fig. 9: Waterfall plot of the UAV Doppler profiles.




(a) Quadcopter F-450




(b) Hexacopter F-550

Fig. 10: Single UAV Doppler profile.

### A. Quadcopter measurements

An example of a single Doppler profile of the F-450 is shown in Fig. 10a. The UAV's body is discernible as the strongest return with a signal-to-noise ratio (SNR) of 24 dB and a Doppler shift of $\mathrm{f_{UAV}} = 7.2\,\mathrm{Hz}$ or a bistatic velocity of 0.19 m/s. Two more returns with much lower SNR are also detectable as peaks at 190.8 Hz and -176.5 Hz respectively. Both returns assume symmetric frequency shifts around $\mathrm{f_{UAV}}$ with deviations of $|\Delta \mathrm{f_{mD}}| = 183.7 \pm 0.1\,\mathrm{Hz}$. With two blades on each rotor, the radar would register two returns from each blade for every full revolution, each once with positive and once with negative Doppler modulation. The translation of this frequency shift to the equivalent rotational speed in revolution-per-minute (RPM) is therefore determined by

$$f_{RPM} = \Delta f_{mD}/2 \cdot 60 \tag{2}$$

which yields $\mathrm{f_{RPM}} = 5511 \pm 3\,\mathrm{RPM}$. This value corresponds well with the lower end of the UAV's specifications in Table II, which seems plausible when considering that the UAV was hovering at this moment with minimal speed of 0.13 m/s.

### B. Hexacopter measurements

Similarly to the quadcopter results, Fig. 10b exemplifies a Doppler-profile taken from measurements with the airborne hexacopter. The strongest return here originates also from the UAV itself with a SNR of 31 dB and its return signal is Doppler-shifted by $\mathrm{f_{UAV}} = 16.7\,\mathrm{Hz}$. As with the smaller quadcopter, two additional Doppler-shifted peaks are visible at much reduced SNR at 190.8 Hz and -221.8 Hz respectively. These returns assume symmetric frequency shifts as well centered around the main return, each separated by $|\Delta \mathrm{f_{mD}}| = 206.3 \pm 0.9\,\mathrm{Hz}$ in Doppler. According to formula (2) this equates to a measured RPM of $\mathrm{f_{RPM}} = 6189 \pm 27\,\mathrm{RPM}$. This value is also consistent with the predicted RPM range reported for this UAV type in Table II. The higher RPM corresponds well with on-board IMU data that logged a higher speed reading of 6.56 m/s.

## VI. Conclusions

These very first results show the feasibility of UAV PBR detection at short ranges by means of using communication satellite signals as IoO. Even with standard processing techniques, the signal CAF already shows no significant ambiguities with a close to ideal thumb-tack response. Ground clutter



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return is also minimal, which makes it easier for the radar to distinguish hovering UAVs from stationary ground clutter.

The appearance of a micro-Doppler signature seems evident. Since exact RPM values were not available, future measurements will log the UAV'S rotational speed in order to validate this finding conclusively. The analyzed signatures of the larger UAV are consistent with the modulations expected it's rotor system. However, detection of these signatures proved to be a very challenging task, since the rotors need to stay at constant RPM during the processing interval in order to coherently integrate returns of the blade flashes. This study only considered the horizontal polarization. Analysis of the potential of dual polarization processing will be subject of further studies.

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